

OptimalControl.jl: A Julia package for modeling and solving optimal control problems with ODEs

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Summary

OptimalControl.jl (Caillau, Cots, Gergaud, Martinon, & Sed, 2024) is a Julia (Bezanson et al., 2017) package for modeling and solving optimal control problems governed by ordinary differential equations (ODEs). As the core of the [control-toolbox ecosystem](#), it provides a unified framework that supports both direct and indirect solution methods with applications spanning aerospace, medical imaging, epidemiology, and quantum control.

The package features an expressive domain-specific language (DSL) built around the `@def` macro, enabling users to define control problems using notation that closely resembles standard mathematical formulations. Problems are solved through direct transcription, converting the continuous problem into a nonlinear program (NLP) using discretization schemes including Euler, trapezoidal, midpoint, and high-order Gauss-Legendre collocation. Alternatively, indirect shooting methods based on Pontryagin's Maximum Principle can be employed. The architecture relies on a modeler-solver separation that provides a modular and extensible foundation, enabling seamless execution on both CPU and GPU with minimal user intervention.

The ecosystem includes extensive [tutorial resources](#), a benchmark problem collection ([OptimalControlProblems.jl](#) (Caillau, Cots, Gergaud, & Martinon, 2024) with formulations in OptimalControl DSL and [JuMP](#) (Lubin et al., 2023)), and performance comparison tools ([CTBenchmarks.jl](#)). Integration with Julia's ecosystem enables access to state-of-the-art tools: NLP solvers [IPOPT](#) (Wächter & Biegler, 2006) and [MadNLP.jl](#) (Shin et al., 2021, 2024), automatic differentiation and NLP modeling through JuliaSmoothOptimizers' [ADNLPModels](#) (Migot et al., 2023), GPU acceleration via [ExaModels.jl](#) (Shin et al., 2023), numerical integration from SciML's [DifferentialEquations.jl](#) (Rackauckas & Nie, 2017), and visualization through [Plots.jl](#) (Christ et al., 2023).

Statement of Need

OptimalControl.jl is, to our knowledge, the only Julia package that unifies both direct and indirect methods for optimal control within a single, coherent framework.

This unified approach addresses a gap in the current optimal control software landscape. Existing tools are fragmented across programming languages and paradigms. Legacy packages such as [HamPath](#) (Caillau et al., 2012), [NutoPy](#) (Caillau et al., 2022), and [COTCOT](#) (Bonnard et al., 2007) implement sophisticated indirect methods but rely on Fortran implementations with MATLAB or Python interfaces, leading to complex and less extensible workflows. Proprietary solvers like [GPOPS-II](#) (Patterson & Rao, 2014) limit transparency and reproducibility. Open-source tools such as [BOCOP](#) (Bonnans et al., 2017), [ACADO](#) (Houska et al., 2011), and

41 [acados](#) ([Verschuere et al., 2021](#)) provide valuable direct method implementations but are not
 42 natively integrated into a modern, high-level scientific computing ecosystem. Python-based
 43 tools such as [Dymos](#) ([Falck et al., 2021](#)), [Pyomo.DAE](#) ([Nicholson et al., 2018](#)), [do-mpc](#) ([Fiedler](#)
 44 [et al., 2023](#)), and [GEKKO](#) ([Beal et al., 2018](#)) offer complementary capabilities but are similarly
 45 limited to direct transcription approaches within the Python ecosystem.

46 Within Julia, most existing packages target specific domains: [RobustAndOptimalControl.jl](#)
 47 for linear systems, [QuantumControl.jl](#) and [Piccolo.jl](#) for quantum optimal control,
 48 [DirectTrajectoryOptimization.jl](#) for trajectory problems, [LinearMPC.jl](#) ([Arnström et al.,](#)
 49 [2022](#)) and [ModelPredictiveControl.jl](#) ([Gagnon et al., 2024](#)) for model predictive control, and
 50 [InfiniteOpt.jl](#) ([Pulsipher et al., 2022](#)) for infinite-dimensional optimization. In contrast with
 51 [InfiniteOpt.jl](#) that is designed as an extension of [JuMP.jl](#), our plan is to provide a modeler
 52 agnostic approach, able to accept general Julia code and leverage various optimization solvers.
 53 Besides, contributing to general-purpose NLP modeling frameworks like JuMP or [CasADi](#)
 54 ([Andersson et al., 2019](#)) would not address the specific needs of optimal control: these tools
 55 lack native support for shooting methods based on Pontryagin's Maximum Principle and that
 56 heavily rely on differential geometric primitives.

57 [OptimalControl.jl](#) fills this gap by providing a DSL that matches mathematical notation, multiple
 58 discretization schemes and shooting methods, with planned support for homotopy continuation
 59 methods. The modeler-solver separation enables complementary use with [InfiniteOpt.jl](#) (JuMP-
 60 based) through alternative NLP modeling backends (currently [ADNLPModels](#), [ExaModels](#))
 61 and solvers. GPU acceleration and an ecosystem with domain-specific applications, tutorials,
 62 and benchmarking tools complete the offering. Target users include researchers and engineers
 63 working in optimal control, control theorists developing new algorithms, and students learning
 64 optimal control through interactive tutorials.

65 State of the Field

66 [OptimalControl.jl](#) requires Julia version 1.10 or later and is registered in the Julia General
 67 registry, enabling straightforward installation via `Pkg.add("OptimalControl")`. We compare
 68 it below with existing software.

- 69 ▪ **Legacy tools (COTCOT, HamPath, NutoPy):** These Fortran packages excel at indirect
 70 methods and homotopy continuation but require multi-language setup (Fortran plus
 71 Matlab / Python). [OptimalControl.jl](#) provides both direct and indirect methods in pure
 72 Julia with straightforward installation via package manager.
- 73 ▪ **Direct method tools (BOCOP, ACADO, GPOPS-II, acados, nosnoc):** Strong direct
 74 method implementations: [GPOPS-II](#) ([Patterson & Rao, 2014](#)) delivers mature methods
 75 with MATLAB and C++ implementations but is proprietary; [acados](#) ([Verschuere et](#)
 76 [al., 2021](#)) targets real-time MPC on embedded systems; [nosnoc](#) ([Nurkanović & Diehl,](#)
 77 [2022](#)) specializes in nonsmooth optimal control; [CasADi](#) ([Andersson et al., 2019](#)), used
 78 as symbolic backend by several of these tools, is a general NLP modeler rather than an
 79 optimal control solver. [OptimalControl.jl](#) offers an open-source alternative with expressive
 80 DSL, native Julia ecosystem integration, GPU support, and unified direct and indirect
 81 approaches.
- 82 ▪ **Julia packages:** [RobustAndOptimalControl.jl](#) targets linear systems; [QuantumControl.jl](#),
 83 [Piccolo.jl](#) and [DirectTrajectoryOptimization.jl](#) serve specific domains; [LinearMPC.jl](#)
 84 ([Arnström et al., 2022](#)) and [ModelPredictiveControl.jl](#) ([Gagnon et al., 2024](#)) focus on
 85 model predictive control. [InfiniteOpt.jl](#) addresses a very rich range of problems, including
 86 optimization on PDE's or with chance constraints, focusing on direct transcription
 87 methods. [OptimalControl.jl](#) works seamlessly on both CPU and GPU, and adds
 88 tools to do shooting in a unified framework plus systematic benchmarking through
 89 [OptimalControlProblems.jl](#) and [CTBenchmarks.jl](#).

Illustrative Example

A self-contained illustrative example is provided as a companion repository (Caillau et al., 2026), archived on Zenodo. It combines direct and indirect solution approaches for a constrained energy minimization problem: a direct method on a coarse grid identifies the three-arc structure (unconstrained–constrained–unconstrained) and initializes a shooting method based on Pontryagin’s Maximum Principle, which then converges to arbitrary precision.

Software Design

The package architecture balances expressiveness, performance, and extensibility through modular design. The core is organized across internal packages within the [control-toolbox organization](#), each hosted as an independent Julia package and registered in the General registry: [CTBase.jl](#) defines the core types (optimal control models, solutions, initial guesses) and their accessors; [CTParser.jl](#) implements the `@def` DSL macro; [CTDirect.jl](#) handles direct transcription and NLP interfacing; and [CTFlows.jl](#) provides Hamiltonian flows and shooting for indirect methods. [OptimalControl.jl](#) re-exports these packages as a unified entry point. Contributors interested in a specific functionality can work directly on the relevant sub-package, each of which has its own documentation, tests, and continuous integration. The delegation of specific responsibilities is as follows:

- **DSL parsing:** [MLStyle.jl](#) (Thautwarm & contributors, 2023) enables pattern matching on abstract syntax trees generated from `@def` macro invocations. This design choice separates problem specification from solution methods, and allows a syntax as close as possible to the mathematical description of the problem.
- **Problem models and solutions:** Structured types represent optimal control models, solutions, and initial guesses. Each type provides textual visualization, accessor methods for introspection, and visual plotting capabilities for solutions. Extensible abstractions enable addition of new problem classes without modifying core code.
- **Discretization:** Direct transcription converts continuous problems into NLPs. Supporting multiple discretization schemes (Euler, midpoint, trapezoidal, Gauss-Legendre) required careful abstraction to share transcription logic while allowing scheme-specific implementations.
- **Modelers and solvers:** We chose a clear modeler-solver separation: discretization produces [ADNLPModels](#) (able to deal with arbitrary user defined functions) or [ExaModels](#) (Shin et al., 2023) instances compatible with multiple NLP solvers ([IPOPT](#) via [NLPModelsIpopt.jl](#) (Orban et al., 2023), [Knitro](#) (Byrd et al., 2006), [MadNLP](#), and [Uno](#) (Vanaret & Leyffer, 2026)). These NLP solvers rely on linear solvers such as [MUMPS.jl](#) (Amestoy et al., 2001, 2019; Montoison et al., 2024) for CPU computations or [CUDSS.jl](#) (NVIDIA Corporation, 2025) for GPU acceleration with [MadNLP](#). Automatic differentiation via [ForwardDiff.jl](#) (Revels et al., 2016) and [DifferentiationInterface.jl](#) (Dalle & Hill, 2026) avoids manual derivative coding.
- **Indirect methods:** Hamiltonian flows integrate with [DifferentialEquations.jl](#) to access adaptive stepping, event handling, and multiple ODE solvers without reimplementing. Shooting methods rely on [NonlinearSolve.jl](#) (Pal et al., 2025) or [MINPACK.jl](#) for root-finding. Future extensions will incorporate homotopy continuation methods leveraging bifurcation analysis tools like [BifurcationKit.jl](#) (Veltz, 2020). Currently, indirect methods are limited to Hamiltonian flows, where the feedback control law depends on both state and costate; support for non-Hamiltonian flows (open-loop or state-feedback control laws) is under active development.
- **Testing and quality assurance:** The multi-repository structure is matched by a layered testing strategy. Each sub-package is tested independently, combining unit tests,

integration tests, and code-quality checks, while `OptimalControl.jl` adds strong end-to-end integration tests that solve complete optimal control problems by both direct and indirect methods. Continuous integration runs across Linux, macOS, and Windows, on both CPU and GPU (through a self-hosted CUDA runner), using reusable GitHub Actions workflows centralized in [CTActions](#) and shared across the ecosystem. Code coverage is tracked on Codecov, downstream packages and applications are guarded against regressions through dedicated breakage tests triggered on pull requests, and beta versions are distributed during development via a local registry, [ct-registry](#). Non-regression on complex problems is increasingly driven by [OptimalControlProblems.jl](#), used as a curated problem bank.

Key design trade-offs:

- DSL vs. programmatic API:** We prioritized a mathematical DSL to reduce cognitive load for domain experts, accepting increased parsing complexity handled internally. Currently we do not address optimization of PDE or stochastic systems.
- GPU acceleration strategy:** The modeler-solver separation enables seamless CPU-to-GPU transition. Users simply select the appropriate modeler-solver pair: `ExaModels` with `MadNLP` and the `CUDSS` linear solver for GPU execution. This modular approach minimizes maintenance burden while enabling GPU performance without reimplementing transcription logic. As for now, GPU support is limited to the `ExaModels` + `MadNLP` modeler-solver combination.
- Method coverage:** Supporting both direct and indirect approaches increases code complexity but serves distinct user needs: direct methods for constrained problems with many variables, indirect methods for theoretical analysis and smaller problems requiring high accuracy. Both approaches resort to iterative solvers and may converge or not, *e.g.* depending on the initial guess. In the case of optimization solvers, the full output status of the solver is returned allowing *a posteriori* analysis.

Research Impact Statement

`OptimalControl.jl` has been applied in published research across multiple domains. Independent external adoption demonstrates the package's broader impact: Ferde et al. (2025) used it for drone racing trajectory optimization, Caio & Silva (2025) for epidemiological models using indirect methods, Morsky (2025) for vaccination strategies under social norms, Opmeer (2025) for optimal harvesting of age-structured populations, and Evangelakos et al. (2025) for fast charging protocols for quantum batteries.

Research by close collaborators further demonstrates the package's versatility across aerospace (Herasimenka et al. (2026)), epidemiology (Bliman et al. (2026)), quantum control (Beschastnyi et al. (2025)), microbial growth control (Innerarity Imizcoz et al. (2025)), and theoretical optimal control (Bouali et al. (2025); Lutz & Privat (2025); Bayen et al. (2026); Bonnard et al. (2026)).

The [control-toolbox organization](#) hosts domain-specific application packages built on `OptimalControl.jl`: medical imaging optimization ([MagneticResonanceImaging.jl](#)), gene regulatory networks ([PWLdynamics.jl](#)), spacecraft orbital transfers ([Kepler.jl](#)), epidemiological modeling ([SIRcontrol.jl](#)), and variational calculus ([CalculusOfVariations.jl](#)).

The package serves educational purposes through [Tutorials.jl](#) covering topics from linear-quadratic regulators to Model Predictive Control. These resources are used in academic courses and workshops. [OptimalControlProblems.jl](#) provides standardized test problems formulated in both `OptimalControl` DSL and `JuMP`, enabling systematic performance comparisons through [CTBenchmarks.jl](#).

Development involves international collaborations with CNES (French space agency), Thales

186 Alenia Space, Inria, and CNRS. The package development began in September 2022, with
187 the first public release in February 2023. As of March 2026, OptimalControl.jl has published
188 more than 40 releases with contributions from multiple developers across the control-toolbox
189 ecosystem packages, demonstrating sustained community engagement.

190 AI Usage Disclosure

191 The core software implementation, algorithms, and architectural design of OptimalControl.jl
192 were developed by the authors. Generative AI has been used to assist with code refactoring
193 and generation of unit and integration tests. This paper was drafted by the authors and
194 subsequently revised with AI assistance for restructuring according to JOSS format requirements
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